CRYPTOVOTE: A BLOCKCHAIN-INSPIRED SECURE E-VOTING SYSTEM

K. Gandhimathi[2], Gokulnath K[1,*], Giridharan P[1], Dhanush K[1]

[1]Department of Computer Science and Engineering,
Muthayammal Engineering College (Autonomous), Rasipuram, Tamil Nadu, India
[2]Department of Computer Science and Engineering,
Muthayammal Engineering College (Autonomous), Rasipuram, Tamil Nadu, India

**Corresponding / Presenting Author Email: gokulnath.k8877@gmail.com*

Abstract

Electronic voting systems frequently rely on centralized architectures that are vulnerable to manipulation, lack transparency, and reduce voter trust. This paper presents CryptoVote, a blockchain-inspired electronic voting system designed to provide secure, tamper-resistant, and auditable elections without the computational overhead of fully decentralized blockchain platforms. Each vote is recorded as a cryptographically hashed block using the SHA-256 algorithm and sequentially linked to previous votes, forming an immutable chain that immediately reveals unauthorized modifications. The system employs a Python-based FastAPI backend and a lightweight JSON ledger for efficient storage, while a web-based frontend enables authentication, vote casting, and verification. Experimental deployment in an institutional election setting demonstrates reliable vote recording, transparent result generation, and efficient auditability.

Keywords: Electronic Voting, Blockchain-Inspired Systems, SHA-256, Secure Voting, Digital Governance

1 Introduction

Electronic voting has gained increasing attention due to its potential to improve accessibility, efficiency, and speed in modern elections. However, traditional electronic voting systems often rely on centralized databases that are susceptible to tampering, insider attacks, and single points of failure. These limitations significantly reduce public trust in digital election processes.

Blockchain technology has been proposed as a solution due to its immutability, transparency, and traceability. Fully decentralized blockchain-based voting systems introduce challenges such as high latency, complex consensus mechanisms, scalability issues, and computational overhead, limiting real-world deployment.

This paper proposes CryptoVote, a blockchain-inspired electronic voting system that adopts core blockchain principles without mining or consensus. The contributions are:

- A lightweight blockchain-inspired voting architecture
- Secure vote recording using cryptographic hash chaining
- Prototype validation in an institutional election environment

2 Related Work

Several studies explore blockchain-based voting systems, emphasizing transparency and immutability while reporting scalability and performance issues. Other approaches rely on decentralized architectures that introduce communication overhead and usability challenges.

CryptoVote differs by eliminating mining and distributed consensus while preserving immutability through cryptographic chaining, enabling improved efficiency.

3 System Architecture

The CryptoVote system comprises frontend, backend, blockchain engine, and storage components. The architecture supports secure vote processing and immutable data storage.

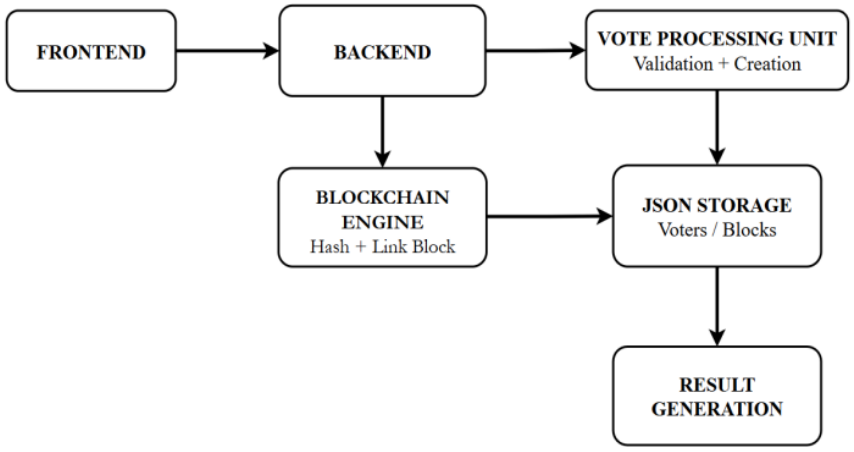


Figure 1: Overall Architecture of the CryptoVote System

As shown in Figure 1, the frontend manages user interaction, the backend validates requests, the blockchain engine ensures immutability, and the storage layer maintains vote records.

4 Methodology

The CryptoVote workflow defines structured interactions between voters and administrators.

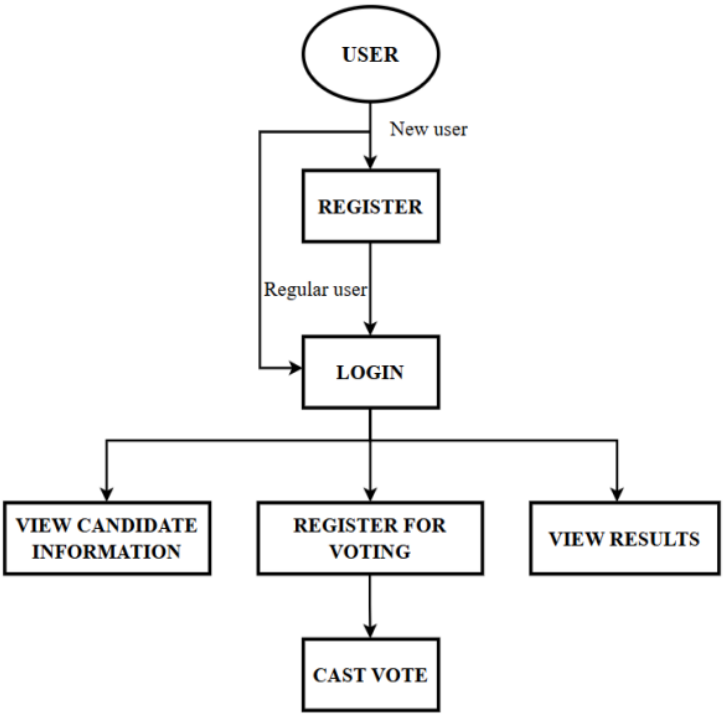


Figure 2: Use Case Diagram of the CryptoVote System

As illustrated in Figure 2, voters authenticate, cast votes, and verify submissions, while administrators manage candidates and results.

The voting process begins with authentication, followed by vote submission and backend validation. Each valid vote is hashed using SHA-256 and linked to the previous block, ensuring tamper detection.

All vote blocks are stored in a JSON-based ledger, providing efficient access and auditability.

5 Implementation

CryptoVote is implemented using Python and FastAPI for backend services. RESTful APIs handle communication between frontend and backend. The frontend uses HTML, CSS, and JavaScript for usability and responsiveness.

6 Results and Discussion

Institutional test elections confirmed accurate vote recording, consistent hash linkage, and transparent result generation. Hash recalculation verified immediate detection of any data modification.

The lightweight design reduced processing overhead compared to fully decentralized blockchain systems while maintaining essential security properties.

7 Conclusion and Future Work

CryptoVote demonstrates that blockchain-inspired principles can enhance electronic voting security without the complexity of full decentralization. Future work includes scalability improvements, biometric authentication, and hybrid trust models.

References

[1] A. Shaikh et al., "Blockchain-Enhanced Electoral Integrity," IEEE Conf. on Secure Digital Governance, 2025.

[2] A. Sharma et al., "Blockchain-Based E-Voting," IEEE Conf. on Smart Digital Systems, 2022.

[3] S. H. Shakir et al., "Fast and Secure Blockchain Voting," IEEE Conf. on Emerging Computing Technologies, 2024.

[4] U. Jafar and M. Ali, "Blockchain for Electronic Voting," IEEE Conf. on Advanced Cyber Technologies, 2021.